

Antonio Manuel

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Professional Summary

Senior Healthcare Data Scientist and Applied AI innovator with 14 years of experience partnering with health plan executives, medical directors, and cross-functional operations to drive business strategy across managed care (Medicaid, and Marketplace). Proven track record pioneering production generative AI solutions, autonomous agentic workflows, and predictive machine learning models that optimize clinical resource allocation, elevate Medicare Star ratings, reduce care gaps, and safeguard payment integrity. Experienced technical leader who bridges complex statistical architectures with commercial and clinical decision-making, leveraging Databricks, PySpark, Python, and Power BI while mentoring multidisciplinary analytics teams to scale enterprise data science capability.

Education

Eastern University, St. Davids, PA, Master of Science (M.S.) in Applied Artificial Intelligence. Relevant Coursework: Deep Learning, Natural Language Processing, Cloud Solutions, Maths for ML, Machine Learning Operations, Generative AI

University of Wisconsin – Milwaukee, Milwaukee, WI, Bachelor of Arts (B.A.) in Actuarial Sciences, Minor in Economics. Relevant Coursework: Regression, Time Series Analysis, Probability Theory, Financial Models, Actuarial Models

Duke University, Durham, NC, Data Science Certificate. Relevant Coursework: Big Data with Hadoop/PySpark, Python for Data Manipulation, Visualization Strategies

Skills

Strategic Healthcare & Managed Care Levers: Medicaid and CHIP Quality Rating System (MAC QRS), Care Gap Reduction, Payment Integrity (FWA), Prior Authorization Automation, Medical Loss Ratio (MLR) Optimization, Cross-Functional Leadership, Executive Consulting

Applied AI & Generative Workflows: Autonomous Agentic Workflows (LangGraph, CrewAI), Retrieval-Augmented Generation (RAG), Fine-Tuning Pre-trained Models (PEFT/LoRA), Clinical NLP, Dense Embeddings, Vector Search (ChromaDB, FAISS), Responsible AI, XAI

Machine Learning & Statistical Methodologies: Supervised & Unsupervised Learning (k-NN, Naive Bayes, SVM, Neural Networks, Decision Trees, Random Forests, XGBoost), Generalized Linear Models (Gamma, Logistic), Causal Econometrics, Time-Series Forecasting

Big Data & Enterprise Cloud Stack: Databricks Delta Lake, Apache Spark, PySpark, Hadoop, Azure AI Studio, AWS (S3, EMR), Linux/BASH, Git, MLOps Drift Monitoring (MLflow)

Data Engineering & Enterprise BI: Advanced SQL Query Optimization, NoSQL, Kimball Dimensional Modeling (Star/Snowflake Schemas), ETL/ELT Automation, Power BI & DAX, Tableau, Interactive HTML Dashboards, Microsoft Office 365 Suite

Professional Experience

Chorus Community Health Plans, Data Scientist (Applied AI/ML & Medical Economics) *Jan 2026 – Present*

- Modernized clinical operations and reduced prior-authorization review latency by 80% by architecting an autonomous agentic workflow (LangGraph) and domain-specific RAG system to parse unstructured coverage determinations for clinical review teams.
- Governed enterprise payment integrity and clinical risk scoring by engineering distributed PySpark data pipelines in Databricks Delta Lake, accelerating analytical turnaround by 45% across multi-million-row longitudinal claims feeds.
- Scaled enterprise analytical capability and established shared ML best practices by founding and leading a monthly technical Think Tank and open office hours for 12 multidisciplinary data scientists and analysts in a matrixed organization.
- Safeguarded model reliability and business continuity across monthly claims adjudication cycles by productionizing automated MLOps pipelines with statistical drift monitoring and CI/CD validation gates in Azure AI Studio.
- Guided strategic decision-making for executive leadership and medical directors by translating complex multivariate model outputs into intuitive, interactive HTML decision tools and executive Power BI dashboards with SHAP feature interpretability.

Chorus Community Health Plans, Senior Data Analyst (Lead ML & Value-Based Strategy) *2019 – 2026*

- Advanced Medicaid and CHIP ratings and closed persistent quality care gaps across 65% of enrolled members by training supervised gradient-boosted ensembles (XGBoost) and deep neural networks on longitudinal diagnostic (ICD-10) and encounter histories.
- Protected health plan financial resources and lifted member medication adherence by 12% by engineering unsupervised anomaly detection models to detect pharmacy billing irregularities, waste, and abuse (FWA) across NDC claims within commercial product.

- Unlocked \$429 PMPM in verified value-based savings and expanded proactive case management referral capture by 22.3% by designing an augmented clinical risk stratification model evaluated via difference-in-differences.
- Secured executive and Board of Directors authorization to hire a dedicated Medical Director by modeling longitudinal utilization patterns, forecasting \$165K–\$380K in monthly savings (\$2,364 PCPM reduction in IOP).
- Validated clinical program ROI and care-gap reduction for executive leadership by executing Gamma GLMs and Monte Carlo permutation testing, proving a statistically significant \$6,903 per-delivery cost avoidance ($p = 0.0176$).
- Isolated \$1.5M in medical expense reductions for 100+ vulnerable families by executing quasi-experimental econometric evaluations and propensity score matching featuring CDC Social Vulnerability Index (SVI) metrics.
- Empowered clinical and operational leadership with self-service decision support by architecting 75% of the enterprise's operational, HEDIS quality, and CHIP ratings dashboards in Power BI and Tableau.
- Unified network contracting and finance operations by transforming fragmented provider reimbursement calculations into an automated, auditable business calculation engine, eliminating multi-week financial reconciliation cycles.

Chorus Community Health Plans, Data Analyst II

2017 – 2019

- Protected health plan premium revenue by improving member retention by 41% within targeted cohort, training and productionizing a supervised logistic regression, SVM, and decision-tree classification model to stratify high-churn cohorts for proactive outreach.
- Increased departmental analytical throughput by 65% by engineering 200+ automated Python and SQL data extraction, outlier cleaning, and feature preprocessing scripts with robust exception-handling logic.
- Standardized member attribution rules, model, and database schema integrity across clinical registries, reducing downstream reporting discrepancies by 35% through cross-functional collaboration with engineering and clinical teams.

Chorus Community Health Plans – Care4Kids, Data Analyst I

2016 – 2017

- Directed clinical data integrity and care-gap closure audits across 5,000+ complex pediatric foster care files, building automated SQL validation pipelines to ensure compliance with state quality mandates.
- Halved recurring operational reporting turnaround from weeks to days by developing automated SQL data pipelines for clinical care coordination teams.

The Benefit Services Group, Data Analyst (Medical Economics Analytics)

2012 – 2016

- Protected commercial account retention and maintained annual claims trend forecasting within 3% accuracy across self-funded employer groups by implementing actuarial rate projection and reserving models, and medical economics benchmarks.
- Identified \$500K+ in actionable healthcare cost containment opportunities for employer clients by evaluating health plan design modifications and modeling longitudinal pharmacy and medical claims.

Industry-Inspired Prototypes

Clinical Policy & Prior Authorization Engine (CPRE)

- Architected an enterprise applied AI solution in Python, PySpark, and Streamlit to automate managed care prior-authorization reviews and patient risk stratification across 116,000+ longitudinal records.
- Implemented an autonomous multi-node LangGraph agentic workflow with dense sentence embeddings and validation gates to evaluate clinical notes against Medicare NCD/LCD policies, generating expedited auto-approvals and structured RFIs.
- Demonstrated an 80% reduction in administrative review latency while maintaining 0.88 ROC-AUC in identifying high-utilization trajectories, translating complex AI outputs into an intuitive clinical decision interface.

PathoLex: Clinical NLP & Diagnostic Entity Extraction

- Engineered an interactive clinical NLP pipeline using transformer classification and spaCy to extract active vs. negated oncological conditions from unstructured EHR notes, preventing diagnostic misclassification.
- Automated the translation of qualitative patient barrier narratives across 5 CMS SDoH domains directly into structured ICD-10 Z-codes, integrating token saliency attribution to give clinical teams transparent model justification.
- Deployed unsupervised BERTopic clustering to discover latent utilization themes across member encounters, structuring clinical insights into interactive Streamlit decision dashboards.

AuraBayes: Probabilistic Financial Risk Engine

- Developed a Bayesian time-series forecasting engine in Python (PyMC) to quantify uncertainty distributions in healthcare expenditure, separating true underlying trend shifts from annual deductible seasonality.
- Delivered an interactive scenario-planning dashboard enabling executive decision-makers to stress-test budget reserve breach probabilities and evaluate risk-bearing contract performance under volatile utilization patterns.